Introduction

CMSC 477 – Robotics: Perception & Planning

Dr. Troi Williams

Slide Credits: Profs. Tokekar and Aloimonos

Special Accommodations

Any student who feels that he or she may need an accommodation because of a disability (learning disability, attention deficit disorder, psychological, physical, etc.), please make an appointment to see me during office hours or email me ASAP.

Instructor

- **Dr. Troi Williams** (Postdoctoral Fellow, CS) [troiw@umd.edu]
- **Anukriti Singh** [anukriti@umd.edu]
  - Office Hours: **TBD**
- **Khuzema Habib** [khabib@umd.edu]
  - Office Hours: **TBD**
- Lab sessions (EAF 3119):
  - **Mon 2:00pm - 4:50pm**
  - **Wed 2:30pm - 5:20pm**

Survey

- How many of you have taken a robotics class or participated in a robotics club before?

Survey

- How many of you have taken a perception class before?

Survey

- How many of you have taken a planning course before?

Survey

- How many of you have worked on a robot before?

What is this class about?

- Robotics is multi-disciplinary, *with fields encompassing*…?
- We will focus on the fundamental problems of planning and perception.
- *What is planning?*
- *What do we mean by perception?*
- *What are these concepts and why are they important???*
- *Let's look at the (tentative) topics!*

How should the robot go from point A to point B?

https://www.willowgarage.com/

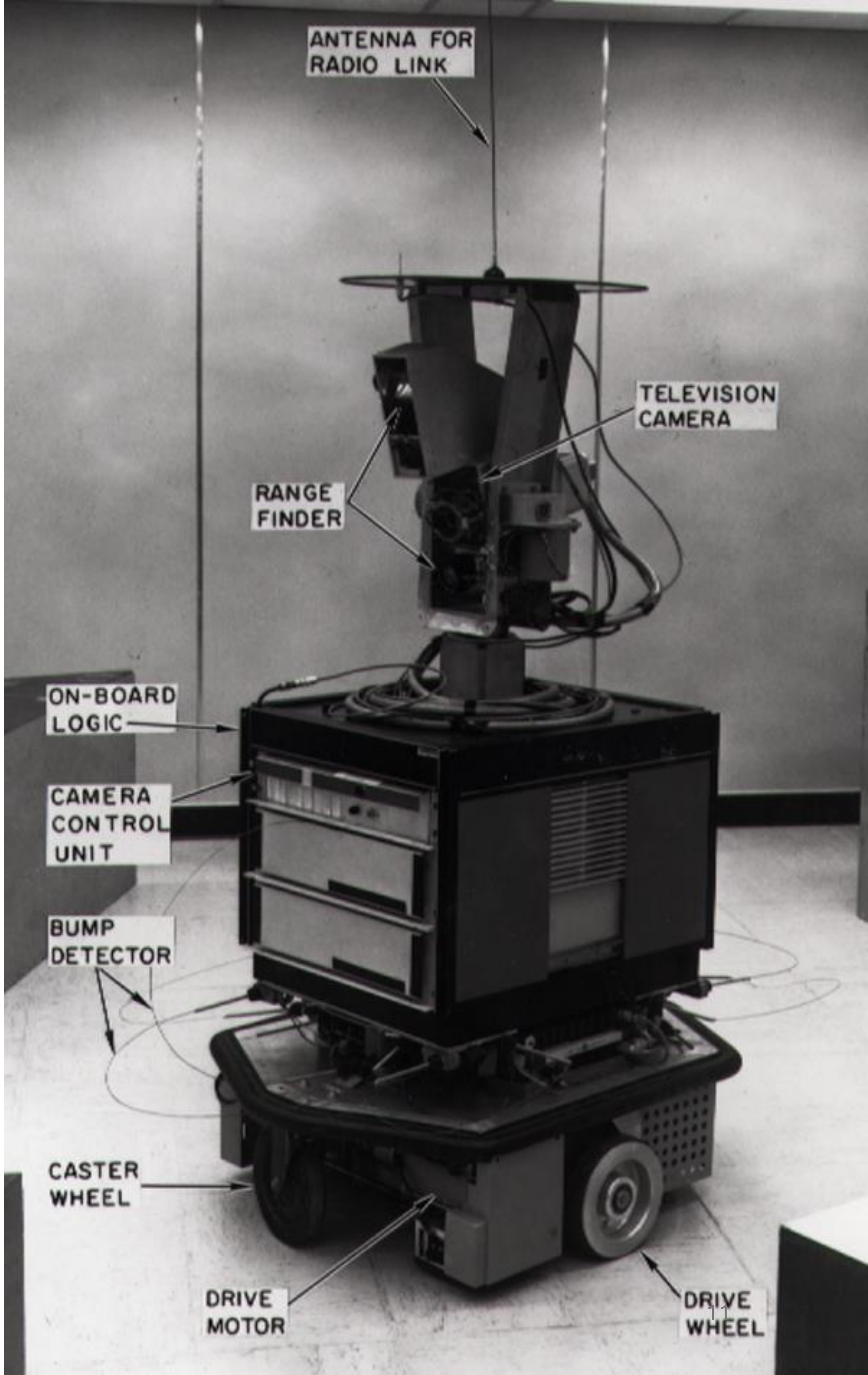
How should the robot go from point A to point B?

- We will start simple (*illustration*):
  - Point robot moving on a graph
  - Known environment with stationary obstacles
  - Perfect sensing
  - Perfect control

- Graph-based path planning
  - Breadth-first search, Depth-first search
  - Dijkstra
  - A*

A*

- Invented by Hart, Nilsson and Raphael of Stanford Research Institute in 1968 for the Shakey robot
- https://youtube.com/clip/UgkxWHPnNZhOKKgzpIGCaonOcwGlsq86X1Dr?si=RM0zkWa94C0LjHGP

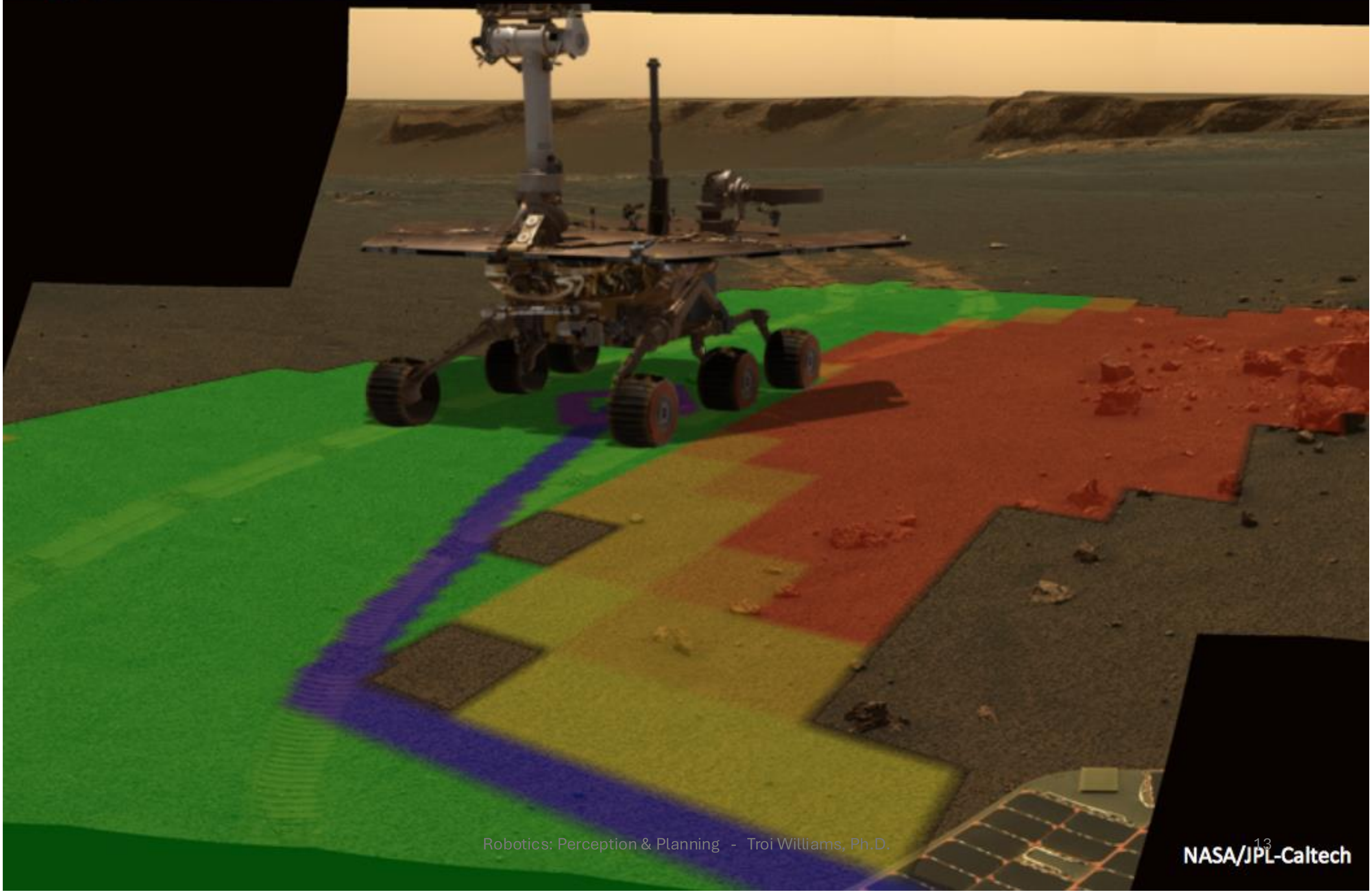


How should the robot go from point A to point B?

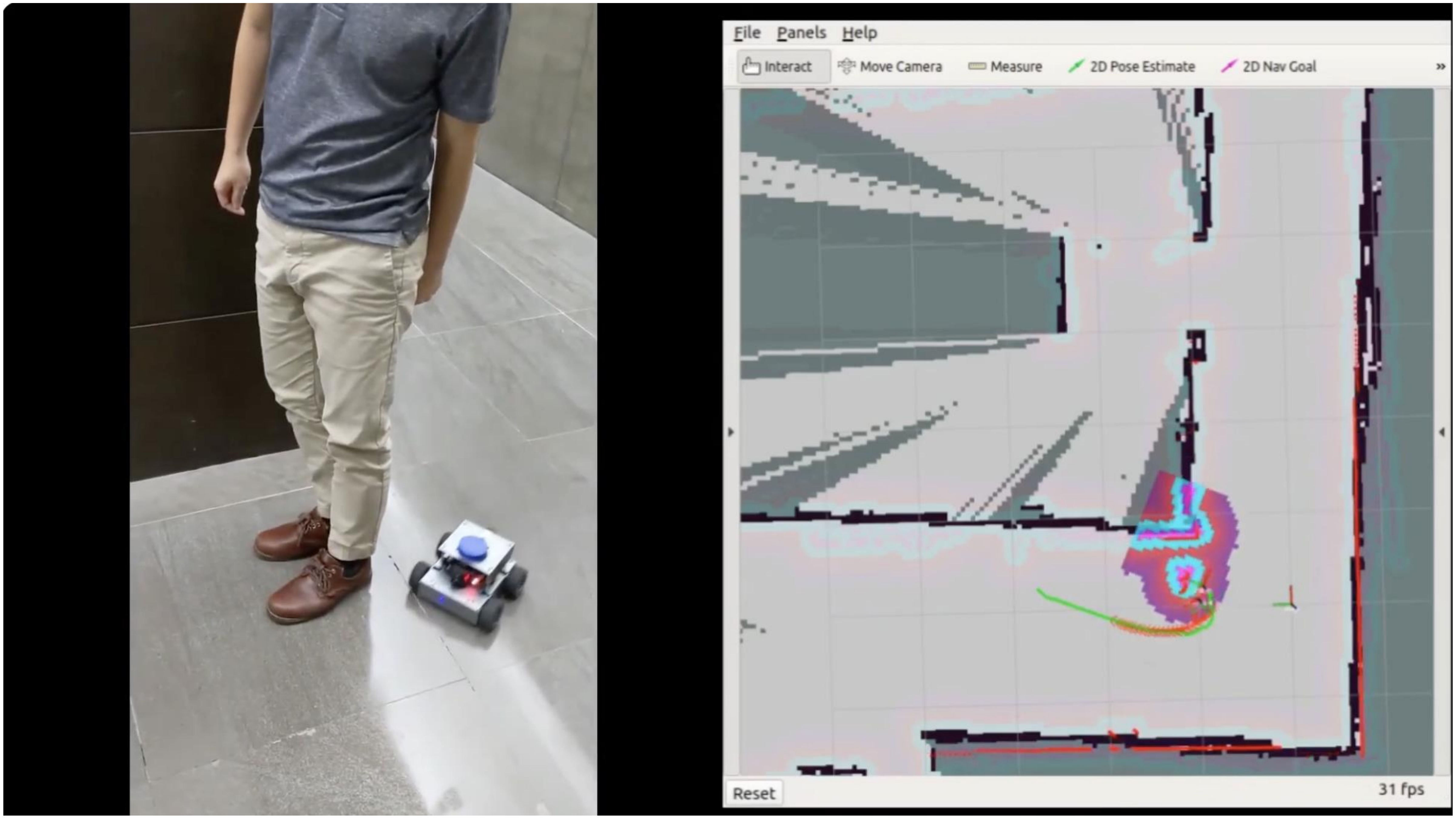
- We will start simple *(illustration):*
  - Point robot that can move in any direction
  - *Unknown environments and/or moving obstacles*
  - Perfect sensing
  - Perfect control

- D* (global) planner
- Dynamic Window Approach (DWA) local planner
- Time Elastic Band (TEB) local planner

D* Global Planner on Opportunity



Time Elastic Band (TEB) local planner



https://www.youtube.com/watch?v=6zoPjC8KV78

How should the robot go from point A to point B?

- We will start simple:
  - Point robot that can move in any direction
  - *Complex, high dimensional environments*
  - Perfect sensing
  - Perfect control

What if I can't discretize the environment?

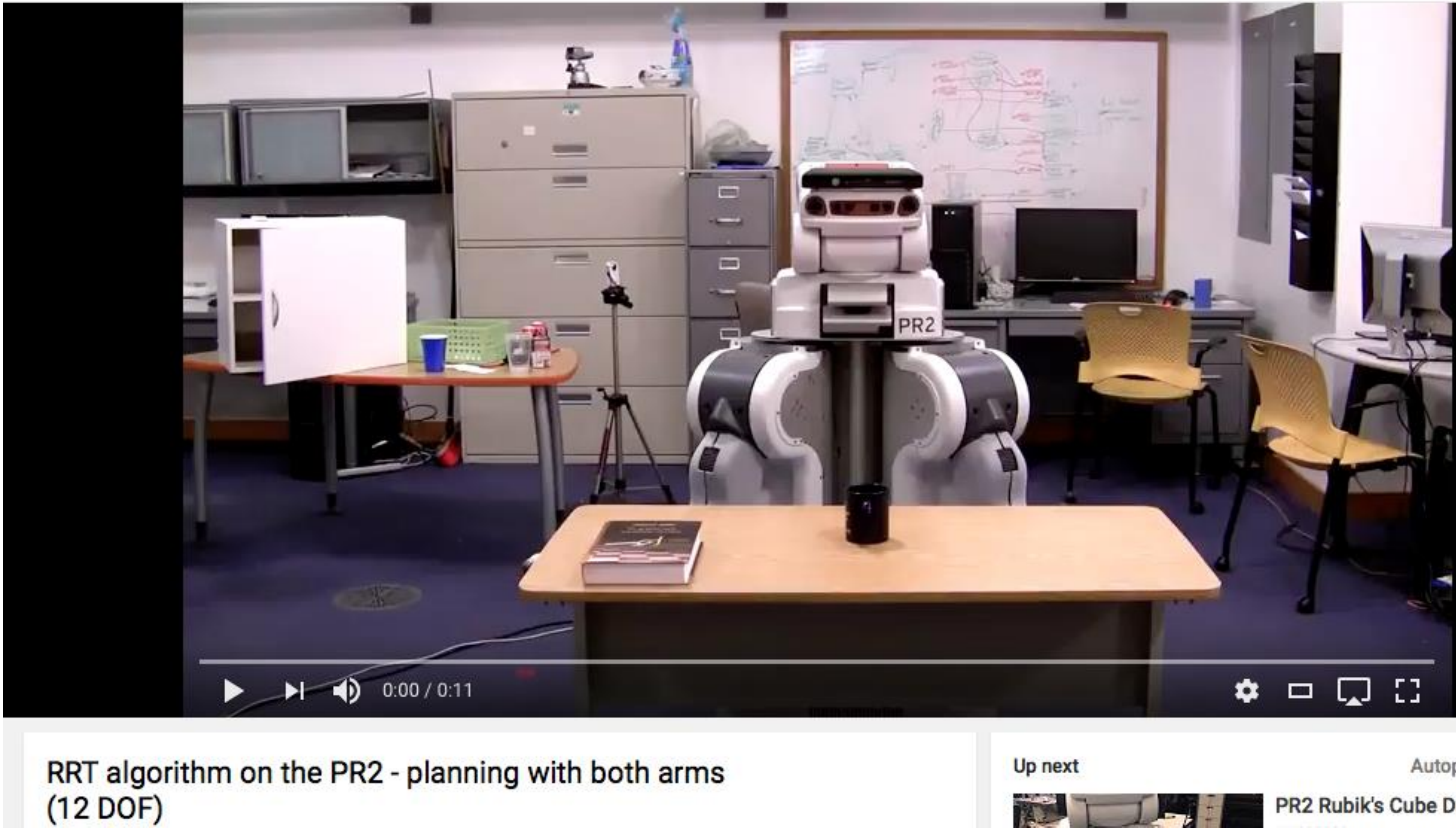
- RRT (sampling-based) planners

Photo: Evan Ackerman

No Tip, Please: nuTonomy's test cars bristle with sensors, are always polite, and in principle should be inexpensive.

The car tries to follow all of the rules all the time, but it breaks the less important ones first: If there's a car idling at the side of the road and partially blocking the lane, nuTonomy's car can break the centerline rule in order to maintain its speed, swerving around the stopped car just as any driver would. The car uses a planning algorithm called RRT*—pronounced "r-r-t-star"—to evaluate many potential paths based on data from the cameras and other sensors. (The algorithm is a variant of RRT, or rapidly exploring random tree.) A single piece of decision-making software evaluates each of those paths and selects the path that best conforms to the rule hierarchy.

http://spectrum.ieee.org/transportation/self-driving/after-mastering-singapores-streets-nutonomys-robotaxis-are-poised-to-take-on-new-cities

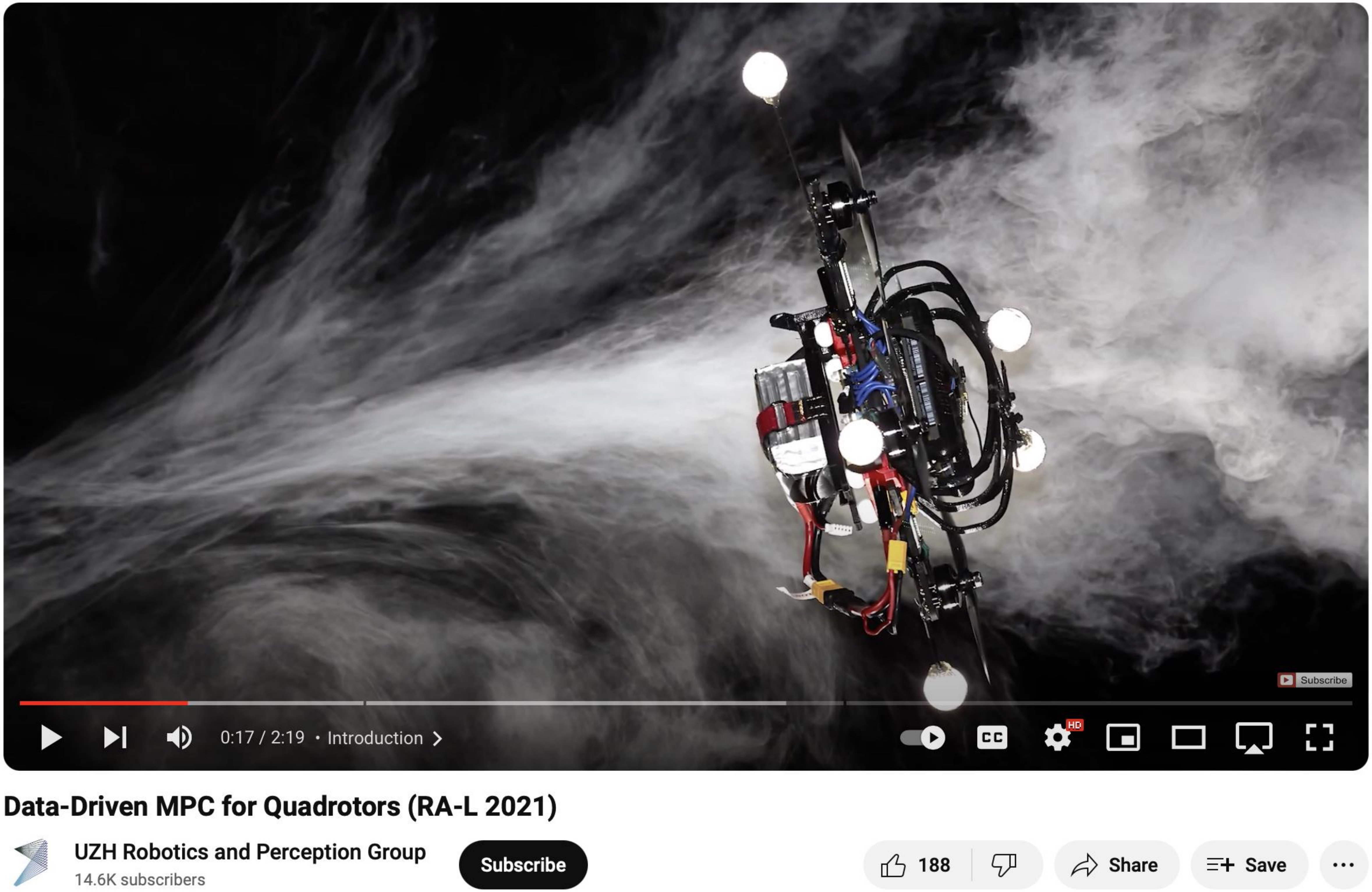


https://www.youtube.com/watch?v=vW74bC-Ygb4

How should the robot go from point A to point B?

- We will start simple:
  - *Realistic robot models*
  - Complex, high dimensional environments
  - Perfect sensing
  - *Imperfect control*

- PID control
- Model Predictive Control (MPC)

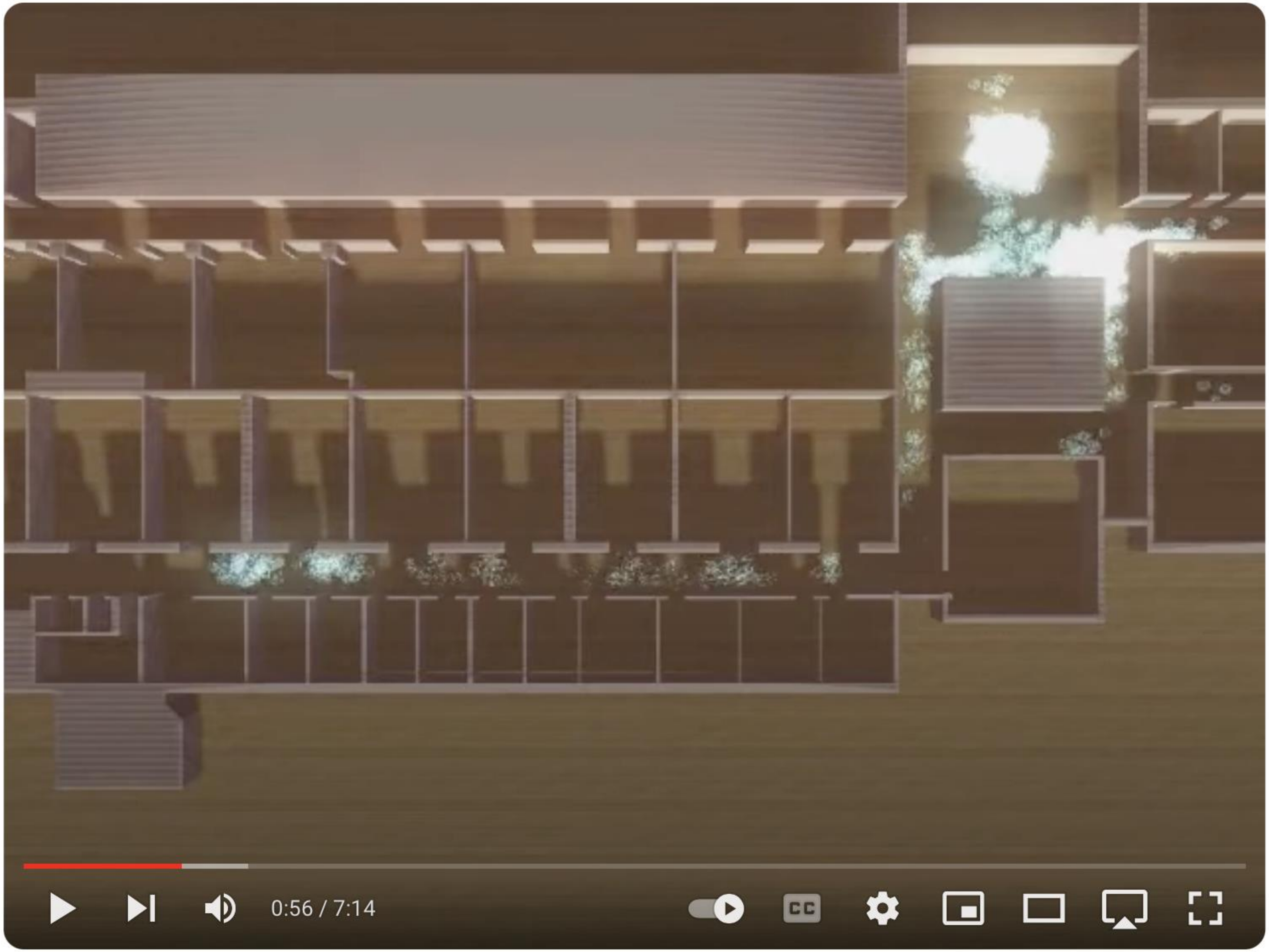


https://www.youtube.com/watch?v=FHvDghUUQtc

How should the robot go from point A to point B?

- We will start simple:
  - Realistic robot models
  - Complex, high dimensional environments
  - *Imperfect sensing*
  - Perfect control

- Kalman Filtering
- Particle Filtering

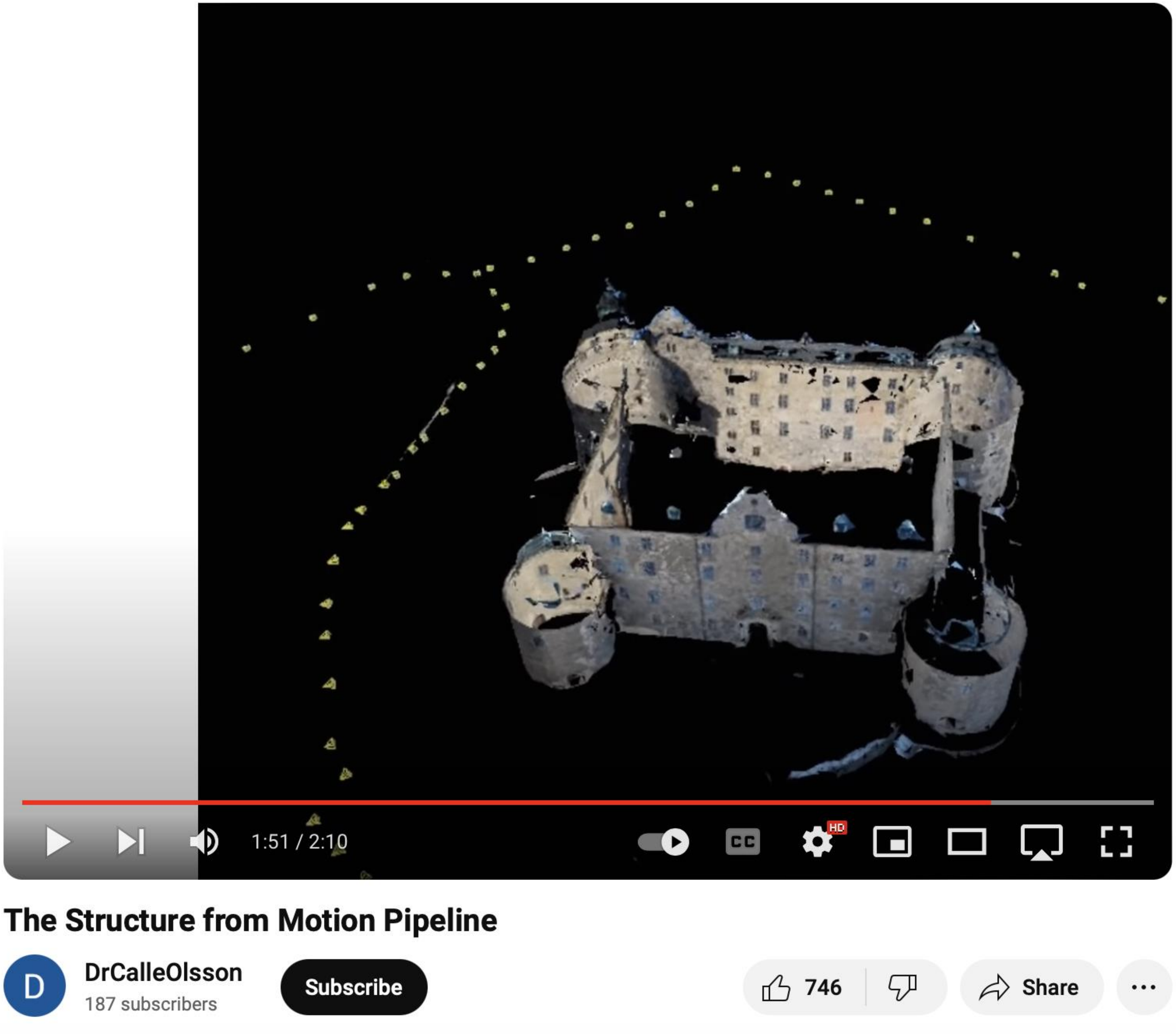


https://www.youtube.com/watch?v=EdHPQ4z6ES8

What about perception?

- Projective Geometry
- Visual Features
- Epipolar Geometry
- Fundamental/Essential Matrix
- Visual Odometry
- Structure from Motion

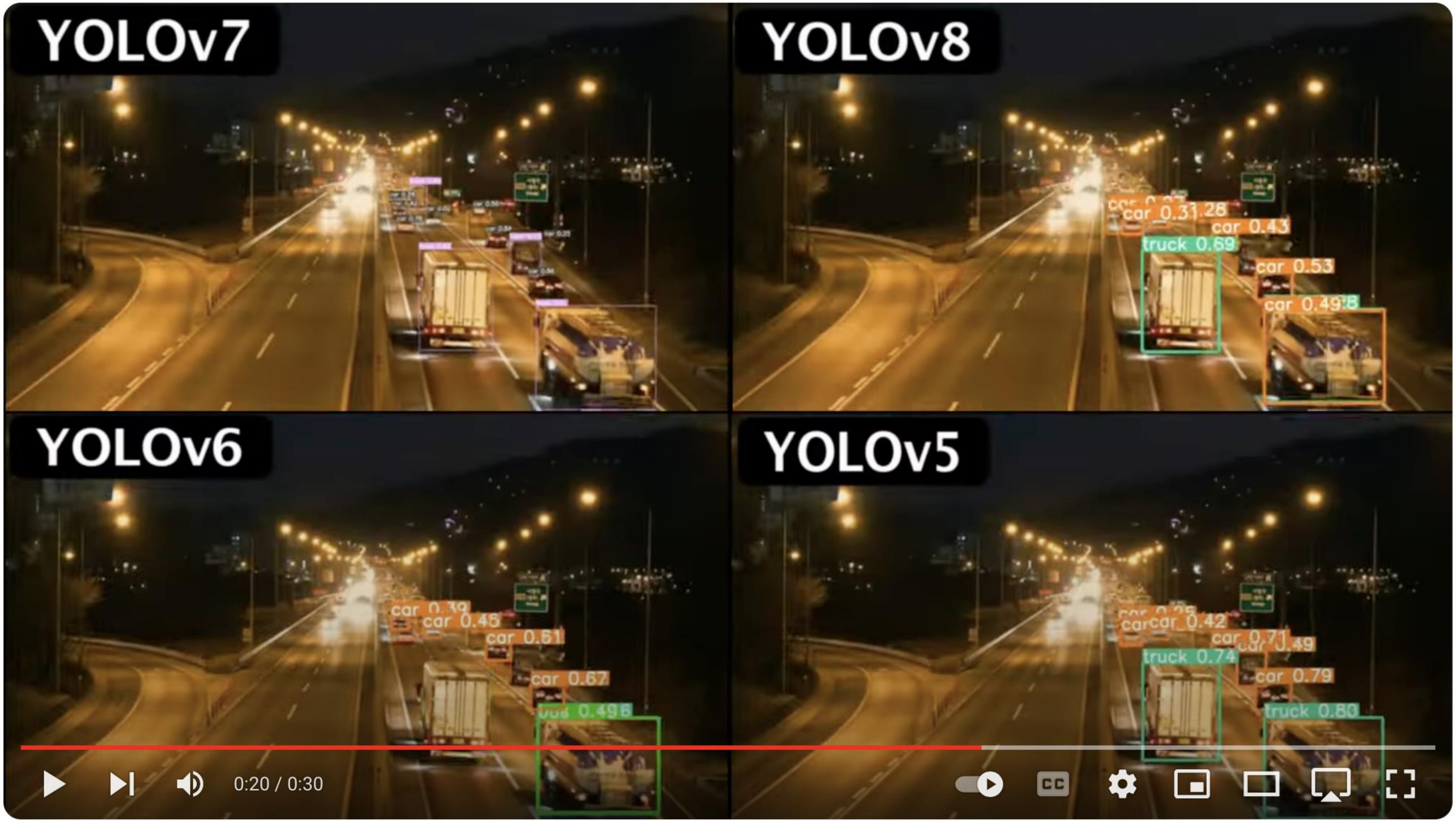
https://www.youtube.com/watch?v=uI8cXy3g8L8



https://www.youtube.com/watch?v=i7ierVkXYa8

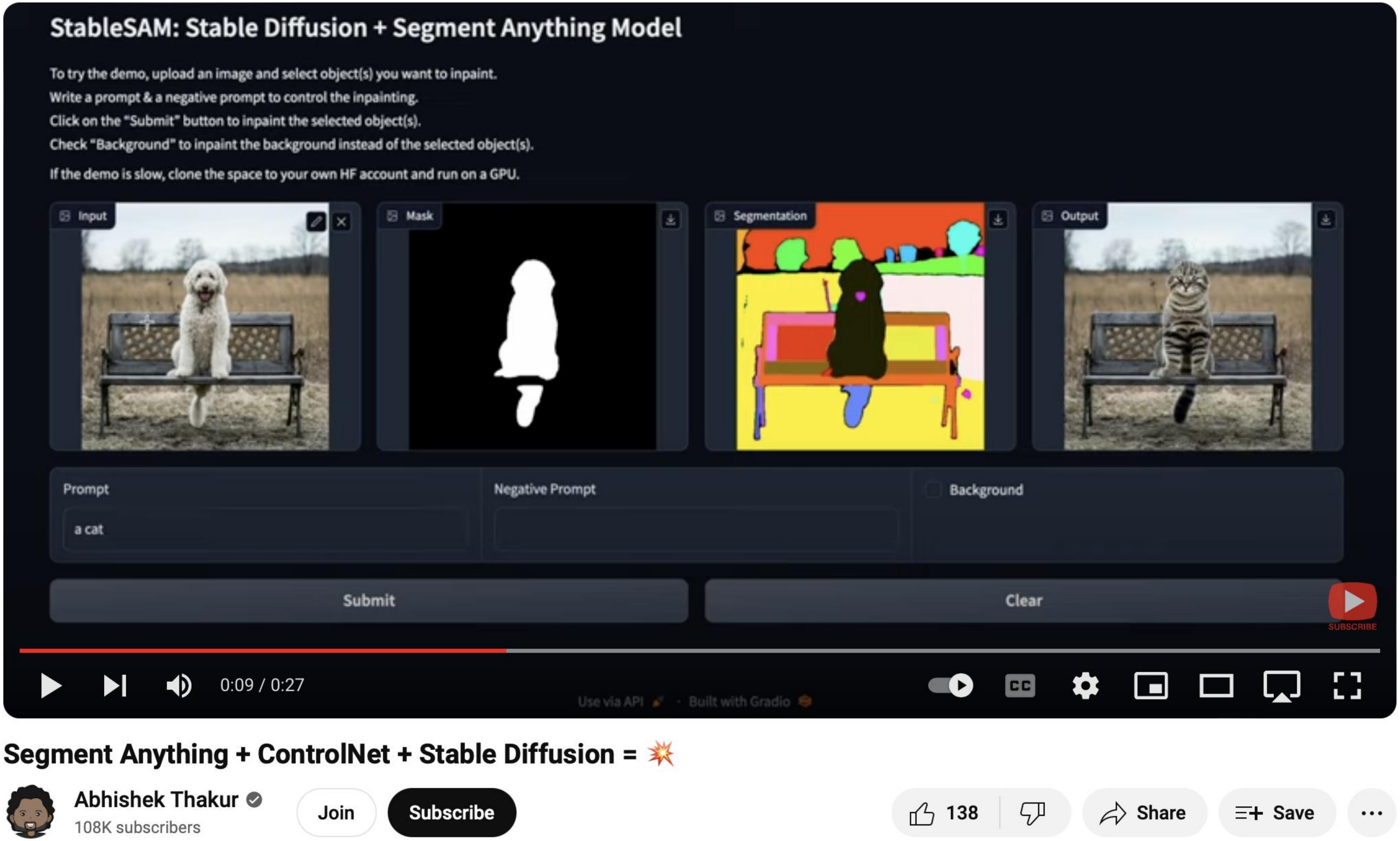
Perception + Learning

- Neural Networks Basics
- Object Detection
- Semantic Segmentation



YOLOv8 Comparison with Latest YOLO models

https://www.youtube.com/watch?v=QOC6vgnWnYo



https://www.youtube.com/watch?v=yIpM4TpAgL4

Structure of the class

- Tues/Thurs lectures (CSI 3117)
- Lab in Idea factory (EAF 3119)
  - **Attend only the one from your section**
  - Section 0101: Monday 2:00pm - 4:50pm (Khuzema Habib)
  - Section 0102: Wednesday 2:30pm - 5:20pm (Anukriti Singh)
- Extra lab sessions will be announced later

Grading*

- 1 Homework (10%)
  - Programming problems

- 4 Projects (65%)
  - Implement on hardware
  - Teams of 2 or 3
  - Will announce logistics next week

- Final Exam (25%)
  - Programming problems

*** subject to change**

Prerequisites

- In addition to official course prerequisite

- Linear Algebra
- Basic data structures
- Comfortable with Python
  - *will be implementing many of the algorithms studied in class*
  - *TA's will help you conceptually, but you must do the coding yourself!*

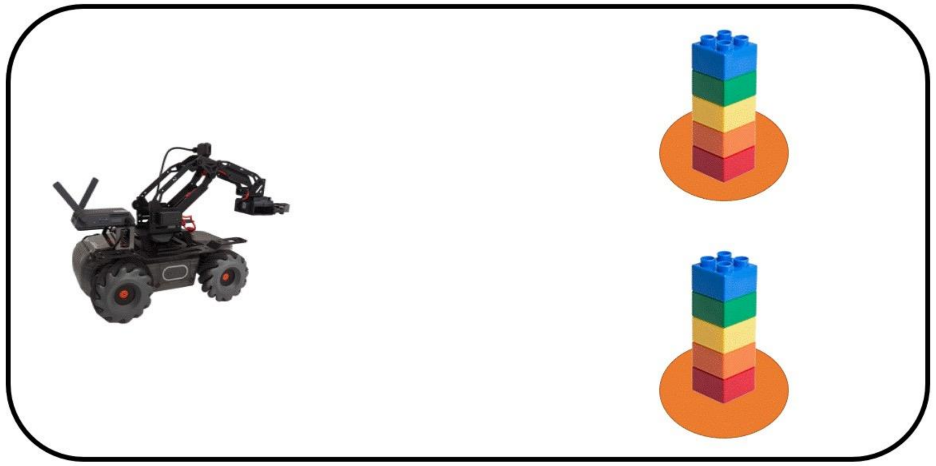
- We will not be using ROS

Projects*

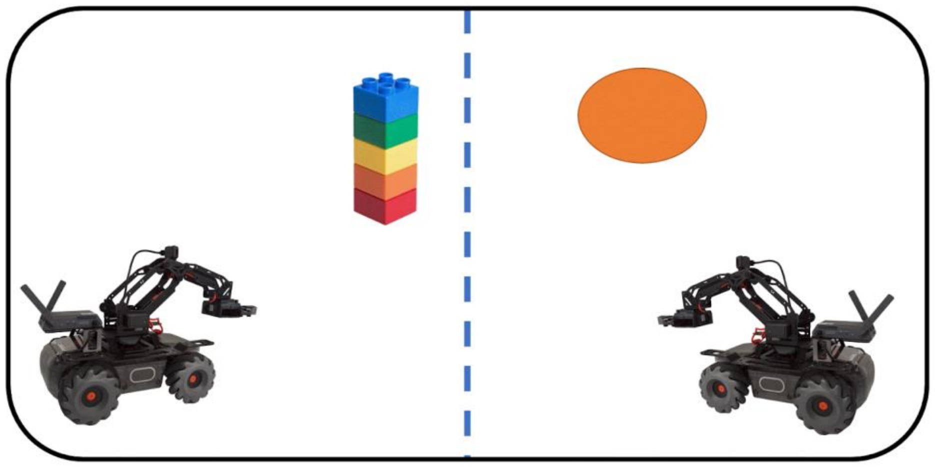
*See info from last year http://prg.cs.umd.edu/cmsc477

Projects

- Project 0: become familiar with hardware
- Project 1: go from point A to B, given the map
- Project 2:



- Project 3:



Spring 2023, CMSC477, final competition

Maryland Robotics Center
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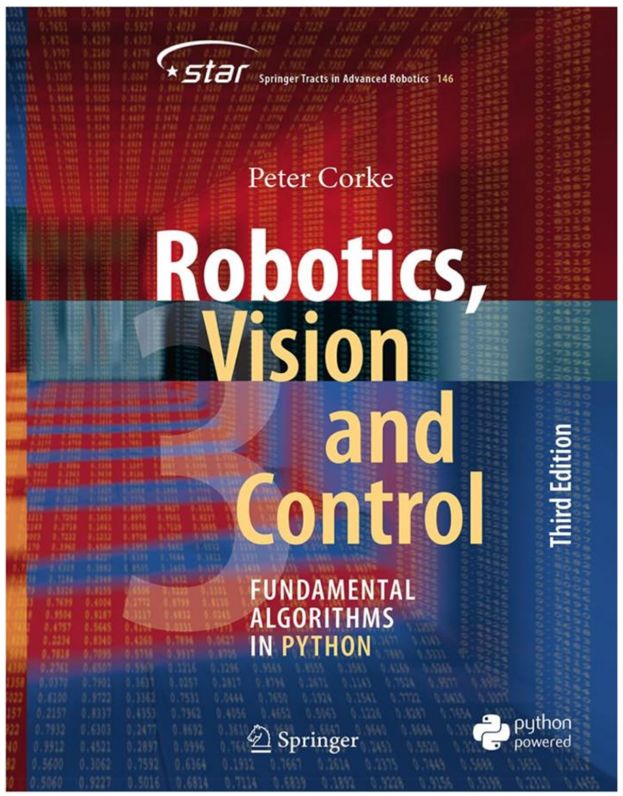
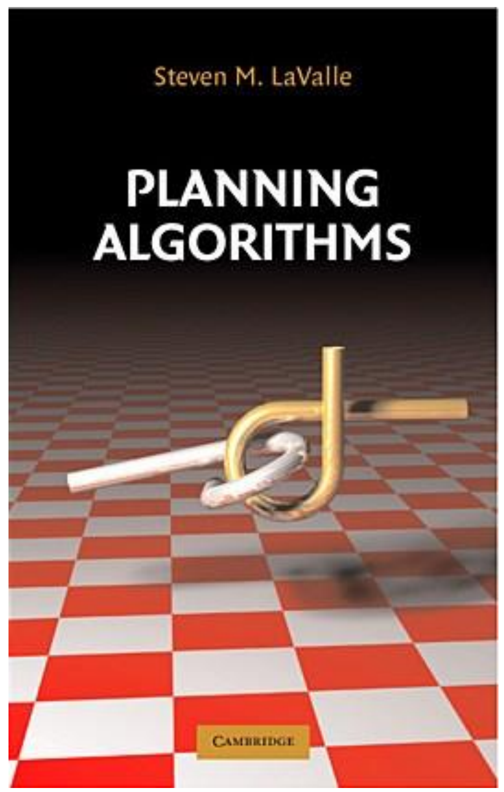
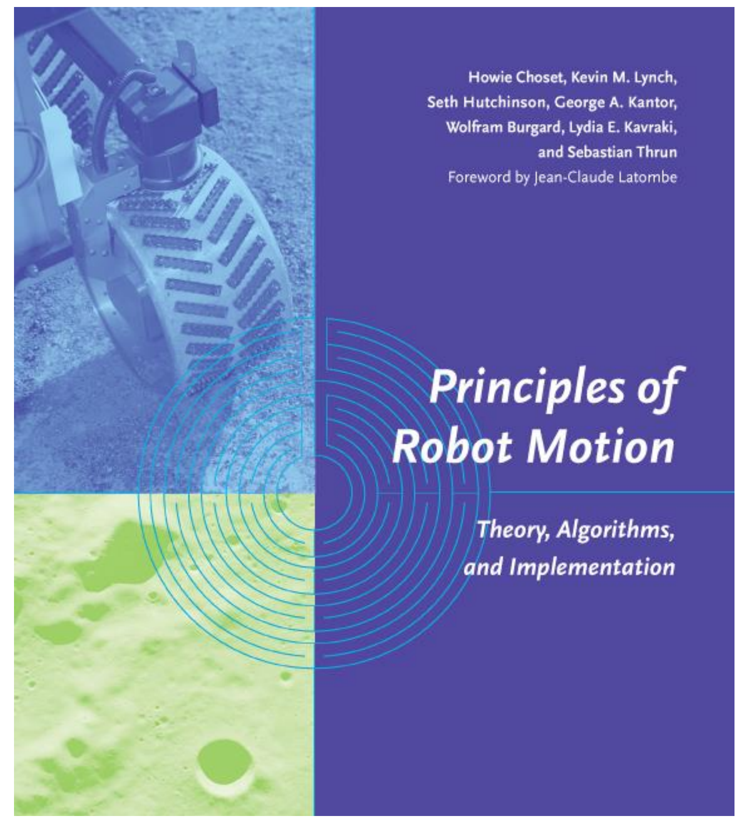
https://www.youtube.com/watch?v=7tRlMvfE3C0

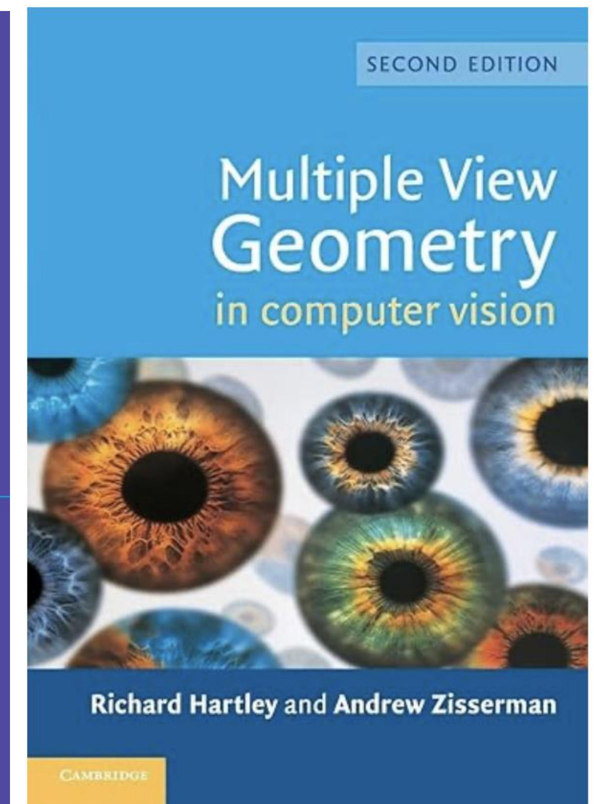
Textbooks

No mandatory textbook but some useful resources









Course Websites

- Syllabus, assignments, schedule, notes and static information (place you need to read):
  - GitHub: https://github.com/troiwill/cmsc477-robotics-perception-and-planning

- Discussion (place where you can write):
  - ELMS: https://umd.instructure.com/courses/1398660

- Make sure to check ELMS and GitHub regularly for updates
- Ensure that you are signed up for notifications

University of Maryland Policies for Undergraduate Students

- Please read the university's guide on Course Related PoliciesLinks to an external site., which provides you with resources and information relevant to your participation in a UMD course.

- Please read Syllabus on GitHub to review the course rules and regulations, *especially the sections related to assignments and using AI.*

Welcome

Let’s get started!